# Online Resource 1: Additional Reproducibility Details

Cong Thanh Bui, Van Anh Nguyen, The Hung Nguyen, Anh Tu Nguyen,
Thi Quynh Le, Thi Linh Lai, and Trung Dung Nguyen

**Article title:** Budgeted Page-Level Preprocessing Selection for Vietnamese Administrative OCR: A Controlled Comparison of Learned, Retrieval-Based, and Static Shortlists
**Journal name:** International Journal on Document Analysis and Recognition
**Corresponding author:** Van Anh Nguyen, National Laboratory for Securing Information (NLSI), Ministry of National Defence, Hanoi, Vietnam. Email: `nguyenanhbh14@gmail.com`

**Caption:** Additional dataset summary, sampled-pool oracle improvement, offline OCR evaluation loop, preprocessing search-space table, surrogate feature groups, nested-CV surrogate model quality, nested GroupKFold protocol details and stepwise algorithm, policy-agnostic headroom controls, Vietnamese best-of-K curves, feature-matched oracle-free selector sensitivity results, SROIE intermediate-budget results omitted from the main manuscript, and contour-suppression schematic referenced from the main manuscript.

## 1 Dataset Summary

Table 1: Dataset summary.

| Aspect | Vietnamese administrative documents | SROIE receipt images |
|---|---|---|
| Dataset/domain | Vietnamese government administrative documents | SROIE receipt images |
| Source | Public Vietnamese Government document portal: `https://chinhphu.vn/he-thong-van-ban` | ICDAR 2019 SROIE benchmark |
| Images | 300 first-page images | 626 images after duplicate removal |
| Time span | 2015–2024, balanced by year | Not applicable |
| Reference text | Author-curated `.docx` transcriptions | Official full-text transcriptions |
| OCR recognizers | VietOCR, Tesseract, EasyOCR | Tesseract, EasyOCR |

## 2 Sampled-Pool Oracle Improvement

Table 2: Sampled-pool oracle improvement over exact NoPre. Relative improvement is computed per page as (NoPre − SampOracle)/NoPre.

| Setting | Recognizer | Pages improved | Mean rel. | Median rel. |
|---|---|---|---|---|
| Raw | VietOCR | 287/300 (96%) | 25.0% | 22.7% |
| Raw | Tesseract | 293/300 (98%) | 35.4% | 36.1% |
| Raw | EasyOCR | 271/300 (90%) | 15.3% | 14.1% |
| Contour | VietOCR | 282/300 (94%) | 27.2% | 25.5% |
| Contour | Tesseract | 287/300 (96%) | 38.2% | 40.5% |
| Contour | EasyOCR | 270/300 (90%) | 16.5% | 15.4% |

# 3 Offline OCR Evaluation Loop

**Input:** Page images and references $\mathcal{D}$; candidate pool $\Theta$; recognizer set $E$; optional contour transform $C$.
**Output:** Collected data table with one row for each page $i$ and preprocessing setting $\theta_j$, containing page features, preprocessing settings, and observed CER values as regression targets.

1. For each page $x_i$, choose the page representation $\tilde{x}_i$: $x_i$ for raw experiments or $C(x_i)$ for contour-suppressed experiments.
2. Extract OCR-free page features $f(\tilde{x}_i)$ once for that page representation.
3. For each candidate $\theta_j \in \Theta$, apply $P_{\theta_j}$ to $\tilde{x}_i$.
4. For each recognizer $e \in E$, run the shared-detection OCR pipeline and compute $y_{i,j,e}$ by page-level CER against $r_i$.
5. Store $z_{i,j}$, $\theta_j$, page/fold identifiers, and CER targets $y_{i,j,e}$.

# 4 Preprocessing Search Space

Table 3: Page-level preprocessing search space.

| Parameter | Range | Off value | Role |
|---|---|---|---|
| `gaussian_ksize` | {0, 3, 5, 7} | 0 | denoising |
| `bilateral_d` | {0, . . . , 9} | 0 | edge-preserving denoising |
| `clahe_clip` | [0, 4.0] | 0 | local contrast enhancement |
| `gamma` | [0.5, 1.5] | 1.0 | brightness correction |
| `sharpen` | [0, 2.0] | 0 | sharpening strength |

# 5 Surrogate Feature Groups

Table 4: Surrogate feature groups.

| Feature group | Features | Purpose |
|---|---|---|
| Base image features | blur, contrast, brightness, noise, edge density, entropy, dynamic range | capture low-level image quality, tone, and texture |
| Document-aware features | text density, skew angle, connected-component count, aspect ratio, estimated line count | capture page structure not represented by intensity statistics alone |
| Preprocessing parameters | `gaussian_ksize`, `bilateral_d`, `clahe_clip`, `gamma`, `sharpen` | describe the candidate preprocessing setting being scored |
| Interaction terms | blur × `gaussian_ksize`, contrast × `clahe_clip`, brightness × `gamma`, noise × `bilateral_d`, entropy × `sharpen` | encode predefined relationships between page properties and preprocessing controls |

# 6 Nested-CV Surrogate Model Quality

# 7 Nested GroupKFold Protocol

# 8 Nested GroupKFold Evaluation Algorithm

**Input:** Collected data table; candidate pool $\Theta$; recognizer $e$; $K$ values $\{3, 5, 10, 15\}$; model-family set $\mathcal{H}$.

Table 5: Nested-CV surrogate model quality. Spearman values are reported as mean ± standard deviation across outer folds. The final refit model is selected and fitted on all available pages for saved inference artifacts; reported metrics are out-of-fold nested-CV results.

| Setting | Recognizer | Final refit model | MAE | $R^2$ | Global Spearman | Per-image Spearman |
|---|---|---|---|---|---|---|
| Raw | VietOCR | Random Forest | 0.021 | 0.463 | $0.592 \pm 0.123$ | $0.629 \pm 0.029$ |
| Raw | Tesseract | XGBoost | 0.046 | 0.614 | $0.674 \pm 0.089$ | $0.497 \pm 0.027$ |
| Raw | EasyOCR | LightGBM | 0.039 | 0.508 | $0.676 \pm 0.062$ | $0.776 \pm 0.013$ |
| Contour | VietOCR | LightGBM | 0.020 | 0.435 | $0.522 \pm 0.040$ | $0.619 \pm 0.037$ |
| Contour | Tesseract | Random Forest | 0.047 | 0.576 | $0.586 \pm 0.104$ | $0.527 \pm 0.029$ |
| Contour | EasyOCR | XGBoost | 0.037 | 0.540 | $0.668 \pm 0.040$ | $0.768 \pm 0.031$ |

Table 6: Nested GroupKFold evaluation protocol.

| Component | Setting | Purpose |
|---|---|---|
| Outer evaluation split | 5-fold GroupKFold grouped by page image | prevents rows from the same page appearing in both training and test data |
| Outer test unit | held-out pages, with all 50 candidates kept together | produces out-of-fold page-level evaluation |
| Inner model selection | 5-fold GroupKFold on each outer training split | selects the surrogate model family by validation MAE without using outer test pages |
| Reported aggregation | held-out page-level results combined across the five outer folds | provides one out-of-fold result per page for CER summaries and statistical tests |
| Final refit | model family selected and refit on all available pages | produces saved inference artifacts, not reported cross-validation metrics |

**Output:** Out-of-fold page-level Surrogate K, Random K, FixedTopK, NN-K, FixedBest, NoPre, and sampled-oracle results.

1. Split pages into five outer GroupKFold folds, keeping all 50 candidate rows from the same page in the same fold.
2. For each outer fold, select $h_e \in \mathcal{H}$ by inner GroupKFold validation MAE on the outer-training pages.
3. Refit the selected model on all outer-training rows and score all $\theta_j \in \Theta$ for each held-out page.
4. Compute Surrogate K, Random K, FixedTopK, NN-K, FixedBest, NoPre, and sampled-oracle references using the definitions in the main manuscript.
5. Pool held-out page-level results across the five outer folds.
6. Compute mean CER, bootstrap confidence intervals, and two-sided Wilcoxon signed-rank tests.

# 9 Policy-Agnostic Headroom Controls

For each held-out page and budget, the main manuscript defines FixedTopK-to-oracle headroom as $H = F - O$. The coupled outcomes $F - P$, for policy $P$, share $F$ with $H$ and satisfy $F - P \leq F - O$. We therefore recomputed the correlation for Surrogate, Random, and NN-K and added direct policy-to-policy controls that do not contain FixedTopK. At every budget, all eight dataset/setting/recognizer configurations have significantly positive coupled correlations for all three policies. Table 7 gives the per-budget breakdown of the additional controls.

Across all budgets, the coupled relationship is positive and significant in 32/32 configurations for each policy, while Random has the larger correlation in 30/32 Surrogate-versus-Random comparisons. The corresponding direct Random-minus-Surrogate correlation ranges from $-0.516$ to 0.022, with no significantly positive result. These controls show that headroom is a retrospective measure of available improvement in the sampled pool, not evidence specific to learned ranking or an oracle-free policy-selection rule.

Table 7: Per-budget headroom-control breakdown. Random $r$ > Surrogate $r$ compares the two coupled correlations. The remaining columns count significantly positive Spearman correlations ($p < 0.05$) for direct policy-to-policy gains that do not contain FixedTopK. Each budget contains eight dataset/setting/recognizer configurations.

| $K$ | Coupled positive, each policy | Random $r$ > Surrogate $r$ | Random − Surrogate | Random − NN-K | NN-K − Surrogate |
|---|---|---|---|---|---|
| 3 | 8/8 | 7/8 | 0/8 | 2/8 | 0/8 |
| 5 | 8/8 | 7/8 | 0/8 | 1/8 | 0/8 |
| 10 | 8/8 | 8/8 | 0/8 | 2/8 | 0/8 |
| 15 | 8/8 | 8/8 | 0/8 | 0/8 | 0/8 |

# 10 Vietnamese Best-of-K Curves

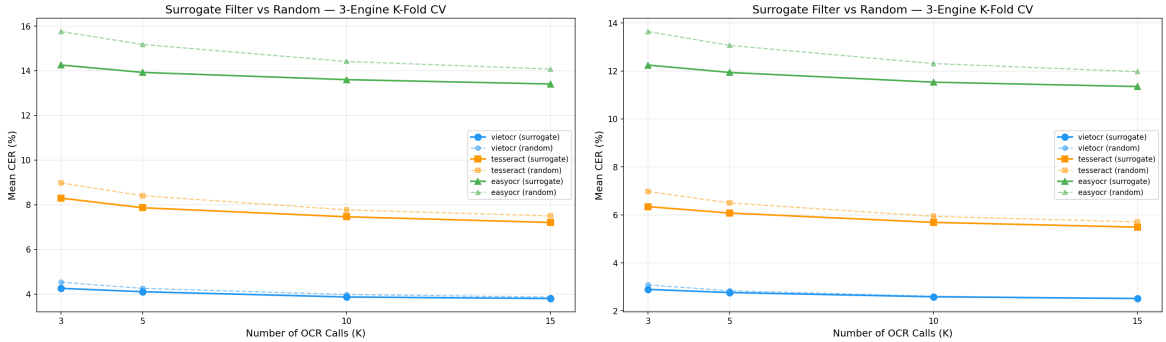



**Fig. 1** Best-of-K CER comparison for raw and contour-suppressed Vietnamese administrative document settings. These curves correspond to the numerical values reported in the main-manuscript Vietnamese best-of-K table

# 11 Feature-Matched Oracle-Free Selector Sensitivity Analysis

The main selector uses continuous surrogate predictions and prediction margins. NN-K and FixedTopK provide only a candidate order. To make these inputs comparable, we replaced the surrogate prediction features by ranks 1, 2, and 3 while preserving the exact surrogate top-3 candidates and OCR outputs. The nested selector was then rerun. Across the eight setting-recognizer combinations, selector CER changed by -0.03 to +0.09 percentage points relative to the full-feature selector. The signs are mixed because the selector is refit, but the small changes are consistent with removing only prediction-related inputs while preserving the exact top-3 candidates and OCR outputs. The table reports paired mean CER differences in percentage points (feature-matched Surrogate minus baseline), percentile bootstrap confidence intervals, and Holm-adjusted two-sided Wilcoxon p-values across the 14 feature-matched comparisons. Negative differences favor the feature-matched Surrogate.

No SROIE NN-K row is included because the stored NN-K shortlist cannot be reconstructed exactly for three EasyOCR pages. The analysis reduces the Surrogate's numerical prediction-feature advantage, but it does not isolate shortlist quality alone because the selector is refit and retains OCR-output features.

# 12 SROIE Intermediate-Budget Results

# 13 Contour-Suppression Schematic

Table 8: Feature-matched oracle-free selector sensitivity analysis. CER values and differences are percentages. Holm p-values are adjusted across all 14 rows in this sensitivity analysis.

| Setting | Recognizer | Policy | Matched Surr. | Policy CER | Diff | 95% CI | Holm $p$ |
|---|---|---|---|---|---|---|---|
| Raw | VietOCR | NN-K | 4.52 | 4.73 | -0.21 | [-0.35, -0.06] | $5.89\times10^{-4}$ |
| Raw | Tesseract | NN-K | 9.03 | 9.31 | -0.28 | [-0.55, -0.02] | 0.433 |
| Raw | EasyOCR | NN-K | 14.99 | 15.33 | -0.34 | [-0.55, -0.13] | 0.0332 |
| Contour | VietOCR | NN-K | 3.15 | 3.27 | -0.12 | [-0.24, 0.00] | 0.458 |
| Contour | Tesseract | NN-K | 7.13 | 7.12 | +0.01 | [-0.26, 0.29] | 1.000 |
| Contour | EasyOCR | NN-K | 12.96 | 13.39 | -0.43 | [-0.64, -0.22] | 0.00125 |
| Raw | VietOCR | FixedTopK | 4.52 | 4.54 | -0.02 | [-0.15, 0.10] | 1.000 |
| Raw | Tesseract | FixedTopK | 9.03 | 9.12 | -0.09 | [-0.36, 0.19] | 1.000 |
| Raw | EasyOCR | FixedTopK | 14.99 | 15.18 | -0.19 | [-0.35, -0.03] | 0.346 |
| Contour | VietOCR | FixedTopK | 3.15 | 3.08 | +0.08 | [-0.02, 0.17] | 0.346 |
| Contour | Tesseract | FixedTopK | 7.13 | 7.05 | +0.08 | [-0.22, 0.39] | 1.000 |
| Contour | EasyOCR | FixedTopK | 12.96 | 13.09 | -0.13 | [-0.31, 0.05] | 0.749 |
| SROIE | Tesseract | FixedTopK | 44.27 | 45.13 | -0.86 | [-1.20, -0.52] | $1.57\times10^{-7}$ |
| SROIE | EasyOCR | FixedTopK | 39.81 | 39.82 | -0.01 | [-0.16, 0.14] | 1.000 |

Table 9: SROIE receipt-setting intermediate-budget CER results. Values are percentages. These rows complement the main-manuscript SROIE table, which reports $K = 3$ and $K = 15$.

| Recognizer | $K$ | Surrogate K | NN-K | FixedTopK | Random K |
|---|---|---|---|---|---|
| Tesseract | 5 | 42.34 | 42.61 | 43.27 | 44.79 |
| Tesseract | 10 | 41.39 | 41.55 | 42.06 | 43.02 |
| EasyOCR | 5 | 38.74 | 38.55 | 38.87 | 39.86 |
| EasyOCR | 10 | 38.32 | 38.27 | 38.46 | 38.88 |

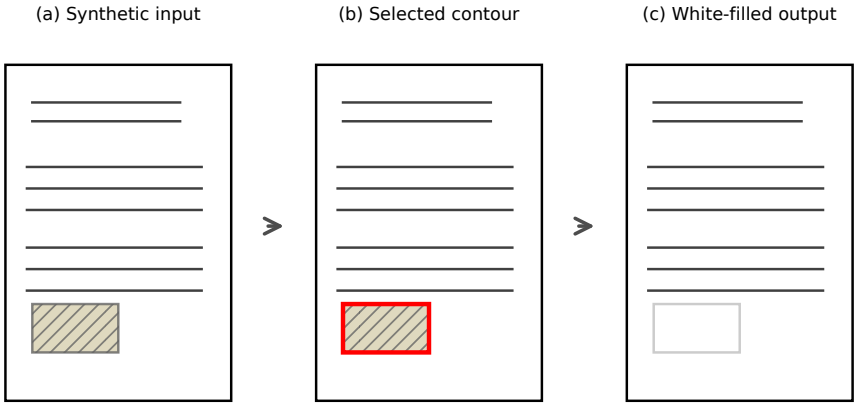



**Fig. 2** Synthetic illustration of contour suppression. A stamp-like rectangular artifact is identified and white-filled before OCR. The figure is schematic and does not reproduce a real source document